THE MAKAR-LIMANOV INVARIANT EQUALS ITS SLICE VERSION IN DIMENSION THREE

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Abstract. Over any field of characteristic zero, the Makar-Limanov invariant of a domain coincides with its slice version whenever the latter has transcendence degree at most one or one less than that of the domain. In dimension three this settles a question of Dasgupta and Gupta, which had been answered only over algebraically closed fields.

Keywords: locally nilpotent derivation; Makar-Limanov invariant; slice.

MSC 2020: 14R10 (primary); 13N15, 13A50, 14R20.

1. Introduction

Let k be a field of characteristic zero and B a commutative k-domain. A k-derivation D of B is *locally nilpotent* if every element of B is killed by some power of D; the set of these derivations is written $\mathrm{LND}(B)$. Makar-Limanov introduced the ring

$$\mathrm{ML}(B) = \bigcap_{D \in \mathrm{LND}(B)} \ker D$$

to prove that the Koras–Russell threefold $x + x^2y + z^2 + t^3 = 0$ is not isomorphic to $\mathbb{C}^3$ although it is contractible [12]. Since then ML has been one of the main tools of affine algebraic geometry. Gupta used a characteristic-free version of it, built from exponential maps, to show that the Zariski cancellation problem has a negative answer for the affine three-space in positive characteristic [10, 11]. The invariant has a known weakness: it can become trivial after adding a variable. Dubouloz proved that the cylinder over the Koras–Russell threefold has trivial Makar-Limanov invariant, the same as the affine four-space [7].

Freudenburg introduced a variant that only uses derivations with a slice [8]. An element $s \in B$ is a *slice* of $D \in \mathrm{LND}(B)$ if $D(s) = 1$. Let $\mathrm{LND}^*(B)$ be the set of locally nilpotent derivations of B that have a slice, and put

$$\mathrm{ML}^*(B) = \bigcap_{D \in \mathrm{LND}^*(B)} \ker D, \qquad \text{with } \mathrm{ML}^*(B) = B \text{ if } \mathrm{LND}^*(B) = \emptyset.$$

If $\mathrm{LND}^*(B) \neq \emptyset$, the intersection defining $\mathrm{ML}(B)$ runs over more derivations, so $\mathrm{ML}(B) \subseteq \mathrm{ML}^*(B)$; if $\mathrm{LND}^*(B) = \emptyset$ the inclusion holds trivially. Dasgupta and Gupta used both rings to characterise the affine plane and the affine three-space [6]. Their main theorem says that, for a three-dimensional affine UFD B over an algebraically closed field of characteristic zero, the following are equivalent: (I) $B = k^{[3]}$; (II) $\mathrm{ML}^*(B) = k$; (III) $\mathrm{ML}(B) = k$ and $\mathrm{ML}^*(B) \neq B$. They showed that the hypotheses "algebraically closed" and "UFD" cannot be dropped from (III)⇒(I), and they asked whether they can be dropped from (III)⇒(II).[1]

Question (Dasgupta–Gupta [6])**.** Does there exist a three-dimensional affine k-domain B over a field k of characteristic zero such that $\mathrm{ML}(B) = k$ but $\mathrm{tr.deg}_k \mathrm{ML}^*(B) = 1$?

[1]The question is numbered 4.8 in the preprint version of [6] and is cited as Question 5.9 in [9]. We quote it verbatim.

They proved that the answer is negative when k is algebraically closed and B is a UFD, and observed that a negative answer in general would give (III)⇒(II) for every three-dimensional affine domain over every field of characteristic zero. Gaifullin and Shafarevich then proved that $\mathrm{ML}(B) = \mathrm{ML}^*(B)$ whenever $\mathrm{LND}^*(B) \neq \emptyset$, for every affine domain over an *algebraically closed* field of characteristic zero [9, Theorem 1]. Their proof compares orbits of the subgroups of $\mathrm{Aut}(B)$ generated by the corresponding unipotent one-parameter groups, through the work of Arzhantsev, Flenner, Kaliman, Kutzschebauch and Zaidenberg on flexible varieties [1]. It needs points over k in general position and so does not apply to fields such as $\mathbb{R}$ or $\mathbb{Q}$. Neither invariant commutes with extension of the base field (Section 5), so the question over non-closed fields cannot be reduced to the closed case.

We answer the question over every field of characteristic zero, without any finiteness or factoriality assumption, by a short argument with no geometry in it.

Theorem A. *Let k be a field of characteristic zero and B a k-domain with $\mathrm{tr.deg}_k \mathrm{ML}^*(B) \leq 1$. Then $\mathrm{ML}(B) = \mathrm{ML}^*(B)$.*

Theorem B. *Let k be a field of characteristic zero and B a k-domain of finite transcendence degree $n \geq 1$ over k with $\mathrm{tr.deg}_k \mathrm{ML}^*(B) = n - 1$. Then $\mathrm{ML}(B) = \mathrm{ML}^*(B) = \ker E$ for every nonzero $E \in \mathrm{LND}(B)$, the ring $\mathrm{ML}^*(B)$ has no nonzero locally nilpotent derivation, and B is a polynomial ring in one variable over $\mathrm{ML}^*(B)$.*

For affine domains Theorem B is a lemma of Dasgupta and Gupta [6, Lemma 3.2], whose proof rests on a theorem of Crachiola and Makar-Limanov on rigid rings [5]. The proof given here does not use that theorem, and it works for every domain of finite transcendence degree. For $\mathrm{tr.deg}_k B \leq 2$ and $\mathrm{LND}^*(B) \neq \emptyset$ the equality $\mathrm{ML}(B) = \mathrm{ML}^*(B)$ was known over every field of characteristic zero [6, 13]; Theorem A is new over fields that are not algebraically closed as soon as $\mathrm{tr.deg}_k B \geq 3$. Together they cover every value of $\mathrm{tr.deg}_k \mathrm{ML}^*(B)$ when $\mathrm{tr.deg}_k B \leq 3$.

Theorem C. *Let k be a field of characteristic zero and B a k-domain with $\mathrm{tr.deg}_k B \leq 3$ and $\mathrm{LND}^*(B) \neq \emptyset$. Then $\mathrm{ML}(B) = \mathrm{ML}^*(B)$.*

Corollary D. *Over a field of characteristic zero there is no domain B of transcendence degree three, affine or not, with $\mathrm{ML}(B) = k$ and $\mathrm{tr.deg}_k \mathrm{ML}^*(B) = 1$. More generally, if $\mathrm{tr.deg}_k B = n$, $\mathrm{ML}(B) = k$ and $k \neq \mathrm{ML}^*(B) \neq B$, then $2 \leq \mathrm{tr.deg}_k \mathrm{ML}^*(B) \leq n - 2$. In particular the conditions (II) and (III) above are equivalent for every three-dimensional domain over every field of characteristic zero.*

The proof rests on one observation, which has nothing to do with the dimension: $\mathrm{ML}^*(B)$ is mapped into itself by *every* locally nilpotent derivation of B, not only by those with a slice (Proposition 3.2). The reason is that the exponential of such a derivation is an automorphism of B, and conjugating a derivation with a slice by an automorphism gives another derivation with a slice. Once this is known, any locally nilpotent derivation E restricts to $\mathrm{ML}^*(B)$, and if the restriction is not zero, a local slice of the restriction can be divided by its image when the kernel of the restriction is small. The quotient is a slice of E itself, so E kills $\mathrm{ML}^*(B)$ after all. This works when the kernel of the restriction is algebraic over k, which is guaranteed by $\mathrm{tr.deg}_k \mathrm{ML}^*(B) \leq 1$. In the case $\mathrm{tr.deg}_k \mathrm{ML}^*(B) = n - 1$ the ring $\mathrm{ML}^*(B)$ is the kernel of a slice derivation, and it has no nonzero locally nilpotent derivations at all.

The remaining range $2 \leq \mathrm{tr.deg}_k \mathrm{ML}^*(B) \leq n - 2$ for $n \geq 4$ is open over non-closed fields; Figure 1 shows what is settled.

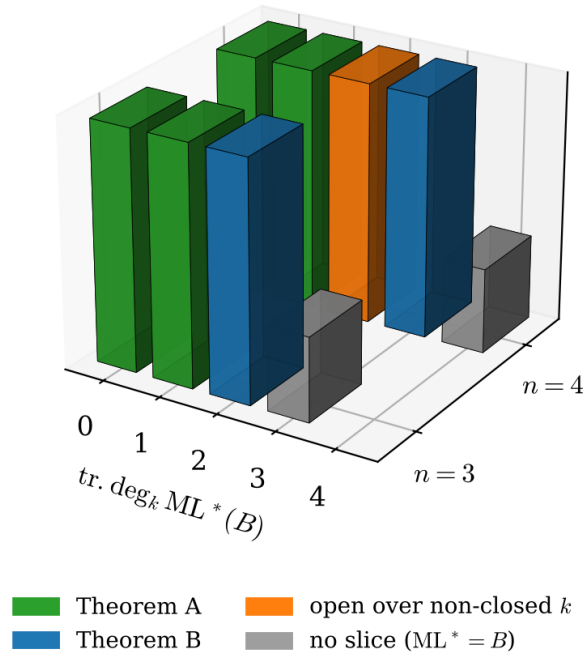


FIGURE 1. Which values of $\operatorname{tr.deg}_k \mathrm{ML}^*(B)$ are settled for $\operatorname{tr.deg}_k B = 3$ and $\operatorname{tr.deg}_k B = 4$ over an arbitrary field of characteristic zero. The value $\operatorname{tr.deg}_k B$ itself occurs exactly when $\mathrm{LND}^*(B) = \emptyset$, where the two rings may differ [6]. Over algebraically closed fields every value is settled by [9].

The paper is self-contained apart from elementary facts on transcendence degree. Section 2 proves the facts on locally nilpotent derivations that we use, including the slice theorem. Section 3 contains the stability of ML^*. Section 4 proves the theorems, and Section 5 discusses examples over the real numbers which show that the two invariants change under extension of scalars.

Conventions. Throughout, k is a field of characteristic zero, rings are commutative with identity, and B is a k-domain, that is, a k-algebra which is an integral domain. We write $\operatorname{tr.deg}_k C$ for the transcendence degree over k of the fraction field of a k-domain C. An element b of B is *algebraic over* a subring C if $P(b) = 0$ for a nonzero polynomial $P \in C[T]$; since denominators can be cleared, this is the same as being algebraic over $\mathrm{Frac}(C)$. A subring $C \subseteq B$ is *algebraically closed in* B if every element of B algebraic over C lies in C. For a ring C, $C^{[1]}$ denotes a polynomial ring in one variable over C. A domain is *rigid* if it has no nonzero locally nilpotent derivation.

2. Locally nilpotent derivations

A k-derivation D of B is k-linear and satisfies $D(ab) = aD(b) + bD(a)$. From $D(1) = D(1 \cdot 1) = 2D(1)$ we get $D(1) = 0$, hence $D(\lambda) = 0$ for $\lambda \in k$, and $\ker D$ is a k-subalgebra of

B. By induction on m one has the Leibniz formula

$$D^m(ab) = \sum_{i=0}^{m} \binom{m}{i} D^i(a)\, D^{m-i}(b). \tag{1}$$

Lemma 2.1. *Let D be a k-derivation of B. The elements of B killed by some power of D form a k-subalgebra. In particular, if B is generated as a k-algebra by elements each killed by a power of D, then $D \in \mathrm{LND}(B)$.*

Proof. The set is closed under sums and multiplication by elements of k, since D^m is k-linear and $D^m(b) = 0$ implies $D^{m'}(b) = 0$ for $m' \geq m$. If $D^m(a) = 0$ and $D^n(b) = 0$, every term of (1) for $D^{m+n}(ab)$ vanishes: for $i \geq m$ the factor $D^i(a)$ is zero, and for $i < m$ the factor $D^{m+n-i}(b)$ is zero because $m + n - i > n$. □

For $D \in \mathrm{LND}(B)$ and $t \in k$ put

$$\exp(tD)(b) = \sum_{i \geq 0} \frac{t^i}{i!} D^i(b),$$

a finite sum for each b. Division by $i!$ is allowed because k has characteristic zero.

Lemma 2.2. *For $D \in \mathrm{LND}(B)$ and $t, u \in k$, the map $\exp(tD)$ is a k-algebra endomorphism of B, and $\exp(tD) \circ \exp(uD) = \exp((t+u)D)$. In particular $\exp(tD)$ is an automorphism with inverse $\exp(-tD)$.*

Proof. Linearity over k and $\exp(tD)(1) = 1$ are clear. By (1),

$$\exp(tD)(ab) = \sum_{m \geq 0} \frac{t^m}{m!} \sum_{i+j=m} \frac{m!}{i!\, j!} D^i(a) D^j(b) = \Big(\sum_i \frac{t^i}{i!} D^i(a)\Big)\Big(\sum_j \frac{t^j}{j!} D^j(b)\Big).$$

Since $\exp(uD)(b)$ is a finite k-linear combination of the $D^j(b)$, all sums below are finite, and

$$\exp(tD)\exp(uD)(b) = \sum_{i,j} \frac{t^i u^j}{i!\, j!} D^{i+j}(b) = \sum_m \frac{1}{m!}\Big(\sum_{i+j=m} \frac{m!}{i!\, j!} t^i u^j\Big) D^m(b) = \exp((t+u)D)(b).$$

Taking $u = -t$ and using $\exp(0 \cdot D) = \mathrm{id}$ gives the inverse. □

The next lemma is the place where characteristic zero enters most directly.

Lemma 2.3. *Let δ be a k-derivation of a k-domain C and $A = \ker \delta$. Every element of C algebraic over A lies in A. In particular $\ker \delta$ is algebraically closed in C, and every element of C algebraic over k lies in $\ker \delta$.*

Proof. Let $b \in C$ be algebraic over A, and choose a nonzero $P = \sum_{i=0}^{m} a_i T^i \in A[T]$ with $P(b) = 0$ and m as small as possible; then $m \geq 1$ and $a_m \neq 0$, since a nonzero constant does not vanish at b. Applying δ to $P(b) = 0$ and using $\delta(a_i) = 0$ gives

$$0 = \sum_{i=1}^{m} i a_i b^{i-1} \delta(b) = P'(b)\, \delta(b), \qquad P' = \sum_{i=1}^{m} i a_i T^{i-1} \in A[T].$$

The leading coefficient $m a_m$ of P' is nonzero because k has characteristic zero and C is a domain, so P' is a nonzero polynomial of degree $m - 1 < m$. By the minimality of m, $P'(b) \neq 0$. As C is a domain, $\delta(b) = 0$. The last assertion follows because $k \subseteq A$. □

Lemma 2.4. *Let δ be a nonzero locally nilpotent derivation of a k-domain C and $A = \ker \delta$. There is $g \in C$ with $c = \delta(g) \neq 0$ and $\delta(c) = 0$. Such an element g is not algebraic over A. Consequently, if $\mathrm{tr.deg}_k C$ is finite, then $\mathrm{tr.deg}_k A \leq \mathrm{tr.deg}_k C - 1$.*

Proof. Pick $f \in C$ with $\delta(f) \neq 0$ and let $N \geq 1$ be the largest integer with $\delta^N(f) \neq 0$; it exists because δ is locally nilpotent. Then $g = \delta^{N-1}(f)$ has $\delta(g) = \delta^N(f) \neq 0$ and $\delta^2(g) = 0$. If g were algebraic over A, Lemma 2.3 would give $\delta(g) = 0$. For the last claim, let $x_1, \ldots, x_r \in A$ be algebraically independent over k. As g is not algebraic over $k[x_1, \ldots, x_r] \subseteq A$, the elements $x_1, \ldots, x_r, g$ of C are algebraically independent, so $r + 1 \leq \operatorname{tr.deg}_k C$. □

An element g as in Lemma 2.4 is called a *local slice* of δ. The next result is the slice theorem; we include the short proof.[2]

Lemma 2.5 (Slice theorem)**.** *Let $D \in \mathrm{LND}(B)$ have a slice s, and let $A = \ker D$. Then s is not algebraic over A, and $B = A[s]$. Thus $B = A^{[1]}$, the derivation D is the derivative d/ds with respect to s, and if $\operatorname{tr.deg}_k B = n$ is finite then $\operatorname{tr.deg}_k A = n - 1$.*

Proof. Since $D(s) = 1 \neq 0$, Lemma 2.3 shows that s is not algebraic over A, so the A-algebra map $A[T] \to B$, $T \mapsto s$, is injective. For $b \in B$ put

$$\pi(b) = \sum_{i \geq 0} \frac{(-1)^i}{i!} D^i(b)\, s^i,$$

a finite sum. Since $D(s) = 1$,

$$D(\pi(b)) = \sum_{i \geq 0} \frac{(-1)^i}{i!} D^{i+1}(b) s^i + \sum_{i \geq 1} \frac{(-1)^i}{(i-1)!} D^i(b) s^{i-1}.$$

After the substitution $i \mapsto i + 1$ the second sum is $\sum_{i \geq 0} \frac{(-1)^{i+1}}{i!} D^{i+1}(b) s^i$, which cancels the first. Hence $\pi(b) \in A$ for every b. Now

$$\sum_{j \geq 0} \frac{1}{j!} \pi\big(D^j(b)\big) s^j = \sum_{i,j \geq 0} \frac{(-1)^i}{i!\, j!} D^{i+j}(b) s^{i+j} = \sum_{m \geq 0} \frac{D^m(b) s^m}{m!} \sum_{i=0}^{m} \binom{m}{i} (-1)^i = b,$$

since the inner sum is $(1-1)^m$, which is 0 for $m \geq 1$ and 1 for $m = 0$. Each $\pi(D^j(b))$ lies in A, so $b \in A[s]$. On $A[s]$ the derivation D kills A and sends s to 1, so it agrees with d/ds. Finally $\mathrm{Frac}(B) = \mathrm{Frac}(A)(s)$ with s transcendental over $\mathrm{Frac}(A)$, so $\operatorname{tr.deg}_k B = \operatorname{tr.deg}_k A + 1$. □

Lemma 2.6. *If $D_1, D_2 \in \mathrm{LND}(B)$ commute, then $D_1 + D_2 \in \mathrm{LND}(B)$.*

Proof. Since $D_1 D_2 = D_2 D_1$, the binomial formula gives $(D_1 + D_2)^N = \sum_{j=0}^{N} \binom{N}{j} D_1^{N-j} D_2^j$. Fix $b \in B$. Choose p with $D_2^p(b) = 0$, and then q with $D_1^q(D_2^j(b)) = 0$ for the finitely many $j < p$. For $N \geq p + q$ every term vanishes: if $j \geq p$ then $D_2^j(b) = 0$, and if $j < p$ then $N - j > q$ and $D_1^{N-j} D_2^j(b) = 0$. □

3. Stability of the slice invariant

From now on write $R = \mathrm{ML}^*(B)$. Each kernel $\ker D$ is a k-subalgebra of B, so R is a k-subalgebra. If $\mathrm{LND}^*(B) \neq \emptyset$, R is algebraically closed in B: an element algebraic over R is algebraic over each $\ker D$, $D \in \mathrm{LND}^*(B)$, so it lies in each of them by Lemma 2.3. The same holds trivially when $R = B$.

Lemma 3.1. *For every k-algebra automorphism σ of B one has $\sigma(R) = R$.*

[2]The map π below is the Dixmier map of s; formally it is $\exp(-sD)$. For the slice theorem in the literature see [8, Chapter 1].

Proof. If $\mathrm{LND}^*(B) = \emptyset$, then $R = B$ and there is nothing to prove. Let $D \in \mathrm{LND}^*(B)$ with slice s, and put $D^\sigma = \sigma D \sigma^{-1}$. It is a k-derivation:

$$D^\sigma(ab) = \sigma\big(\sigma^{-1}(a)D(\sigma^{-1}b) + \sigma^{-1}(b)D(\sigma^{-1}a)\big) = aD^\sigma(b) + bD^\sigma(a).$$

It is locally nilpotent since $(D^\sigma)^m = \sigma D^m \sigma^{-1}$, it has the slice $\sigma(s)$ because $D^\sigma(\sigma(s)) = \sigma(D(s)) = \sigma(1) = 1$, and $\ker D^\sigma = \sigma(\ker D)$. The map $D \mapsto D^\sigma$ is a bijection of $\mathrm{LND}^*(B)$ onto itself, with inverse $D \mapsto D^{\sigma^{-1}}$. Hence

$$\sigma(R) = \bigcap_{D \in \mathrm{LND}^*(B)} \sigma(\ker D) = \bigcap_{D \in \mathrm{LND}^*(B)} \ker D^\sigma = R. \qquad \square$$

Proposition 3.2. *Every $E \in \mathrm{LND}(B)$ satisfies $E(R) \subseteq R$. Hence E restricts to a locally nilpotent derivation $E|_R$ of R.*

Proof. Let $a \in R$ and choose N with $E^{N+1}(a) = 0$. For $t \in k$, Lemmas 2.2 and 3.1 give

$$\exp(tE)(a) = \sum_{i=0}^{N} t^i \, \frac{E^i(a)}{i!} \in R. \tag{2}$$

The field k contains $\mathbb{Q}$ and is infinite, so we may choose distinct $t_0, \dots, t_N \in k$. The Vandermonde matrix $V = (t_j^i)_{0 \le j,i \le N}$ has determinant $\prod_{j<j'}(t_{j'} - t_j) \neq 0$, so it is invertible over k. By (2) the column vector with entries $E^i(a)/i!$ equals V^{-1} applied to the column with entries $\exp(t_j E)(a) \in R$. Its entries are therefore k-linear combinations of elements of R, and lie in R because R is a k-subspace. In particular $E(a) \in R$. The restriction is locally nilpotent because E is. $\square$

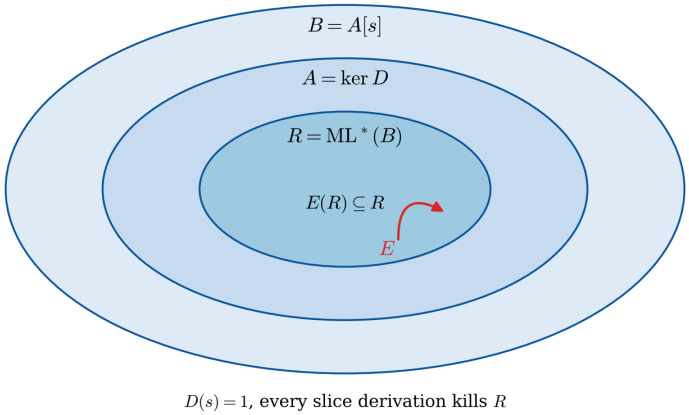


FIGURE 2. The rings in the proofs. A slice derivation D writes B as $A[s]$ with $A = \ker D$, and $R = \mathrm{ML}^*(B)$ lies in every such A. By Proposition 3.2 an arbitrary locally nilpotent derivation E maps R into itself, even when E moves A.

The following extension lemma is [6, Lemma 3.1(i)]; we recall the proof, which we also need for non-affine rings.

Lemma 3.3. *Let $D \in \mathrm{LND}^*(B)$ and $A = \ker D$. Then $R \subseteq \mathrm{ML}(A)$.*

Proof. By Lemma 2.5, $B = A[s]$ is a polynomial ring and $D = d/ds$. Let $\delta \in \mathrm{LND}(A)$ and define $\tilde{\delta}$ on $A[s]$ coefficientwise, $\tilde{\delta}(\sum_i a_i s^i) = \sum_i \delta(a_i)s^i$. It is a k-derivation, since δ is one on A and $\tilde{\delta}(s) = 0$. It is locally nilpotent, because $\tilde{\delta}^m(\sum_i a_i s^i) = \sum_i \delta^m(a_i)s^i$ and only finitely many a_i occur. It commutes with D, since for $a \in A$

$$\tilde{\delta}D(as^i) = \tilde{\delta}(ia\, s^{i-1}) = i\,\delta(a)s^{i-1} = D\big(\delta(a)s^i\big) = D\tilde{\delta}(as^i).$$

By Lemma 2.6, $E = \tilde{\delta} + D$ is locally nilpotent, and $E(s) = 0 + 1 = 1$, so $E \in \mathrm{LND}^*(B)$. For $a \in A$ we have $E(a) = \delta(a)$, so $\ker E \cap A = \ker \delta$. Since $R \subseteq \ker D = A$ and $R \subseteq \ker E$, we get $R \subseteq \ker \delta$. This holds for every $\delta \in \mathrm{LND}(A)$, so $R \subseteq \mathrm{ML}(A)$. □

Remark 3.4. Lemma 3.1 and Proposition 3.2 say that R is stable under the whole automorphism group of B and under every locally nilpotent derivation, although it is defined by a smaller family. The ring $\mathrm{ML}(B)$ is the ring of invariants of the subgroup of $\mathrm{Aut}(B)$ generated by all the groups $\{\exp(tD) : t \in k\}$, and R is the ring of invariants of the subgroup generated by those with $D \in \mathrm{LND}^*(B)$; the second subgroup is normal in the first. Over an algebraically closed field Gaifullin and Shafarevich show that the two subgroups have the same general orbits [9]. Group actions and the structures they preserve in other geometric settings are studied in [2, 3, 4].

4. Proofs of the theorems

Proof of Theorem A. We have $\mathrm{ML}(B) \subseteq R$ (Section 1), so it suffices to show that $R \subseteq \ker E$ for every $E \in \mathrm{LND}(B)$. Fix E and let $\varepsilon = E|_R$, which is a locally nilpotent derivation of the domain R by Proposition 3.2. Suppose, to reach a contradiction, that $\varepsilon \neq 0$, and let $K = \ker \varepsilon \subseteq R$.

By Lemma 2.4 there is $g \in R$ with $c = \varepsilon(g) \neq 0$ and $\varepsilon(c) = 0$, and g is not algebraic over K. We claim that every element of K is algebraic over k. Otherwise some $x \in K$ is transcendental over k; as g is not algebraic over $k[x] \subseteq K$, the elements x and g of R are algebraically independent over k, which contradicts $\mathrm{tr.deg}_k R \leq 1$.

So $c \in K$ is a nonzero element algebraic over k. Let $\mu(T) = T^m + \lambda_{m-1}T^{m-1} + \cdots + \lambda_0 \in k[T]$ be a monic polynomial of least degree with $\mu(c) = 0$. If $\lambda_0 = 0$, then $c \cdot (c^{m-1} + \lambda_{m-1}c^{m-2} + \cdots + \lambda_1) = 0$ with $c \neq 0$ in a domain, which contradicts the minimality of m. Hence $\lambda_0 \neq 0$ and

$$c^{-1} = -\lambda_0^{-1}\big(c^{m-1} + \lambda_{m-1}c^{m-2} + \cdots + \lambda_1\big) \in k[c] \subseteq K.$$

Put $s = c^{-1}g \in R \subseteq B$. Since $c^{-1} \in K$, we have $E(c^{-1}) = \varepsilon(c^{-1}) = 0$, and therefore

$$E(s) = E(c^{-1})\, g + c^{-1}E(g) = c^{-1}c = 1.$$

Thus s is a slice of E, so $E \in \mathrm{LND}^*(B)$, and by the definition of R we get $R \subseteq \ker E$, that is, $\varepsilon = 0$. This contradiction shows $\varepsilon = 0$, so $R \subseteq \ker E$. □

The argument shows a little more: if $E \in \mathrm{LND}(B)$ does not kill $\mathrm{ML}^*(B)$, then $\ker(E|_{\mathrm{ML}^*(B)})$ contains an element transcendental over k.[3] The step that fails in general is the inversion of c: for $\mathrm{tr.deg}_k \mathrm{ML}^*(B) \geq 2$ the element $c = \varepsilon(g)$ may be a non-unit of K, and $c^{-1}g$ is then not in B.

Proof of Theorem B. Since $\mathrm{tr.deg}_k R = n-1 < n = \mathrm{tr.deg}_k B$, we have $R \neq B$, so $\mathrm{LND}^*(B) \neq \emptyset$. Fix $D \in \mathrm{LND}^*(B)$ with slice s and put $A = \ker D$. By Lemma 2.5, $B = A[s] = A^{[1]}$ and $\mathrm{tr.deg}_k A = n - 1$.

[3] When $\mathrm{tr.deg}_k \mathrm{ML}^*(B) = 0$ the theorem is also immediate from Lemma 2.3: every element of $\mathrm{ML}^*(B)$ is then algebraic over k, hence killed by every derivation of B.

Step 1: $R = A$. We have $R \subseteq A$. Choose $x_1, \dots, x_{n-1} \in R$ algebraically independent over k; this is possible because $\mathrm{Frac}(R)$ is generated by R. Let $a \in A$. If a were not algebraic over $k[x_1, \dots, x_{n-1}]$, the n elements $x_1, \dots, x_{n-1}, a$ of A would be algebraically independent, against $\mathrm{tr.deg}_k A = n - 1$. So a is algebraic over R, and $a \in R$ because R is algebraically closed in B (Section 3). Hence $A = R$, and $B = R^{[1]}$.

Step 2: R is rigid. By Lemma 3.3 and Step 1, $R \subseteq \mathrm{ML}(A) = \mathrm{ML}(R)$. So every $\delta \in \mathrm{LND}(R)$ vanishes on R, that is, $\mathrm{LND}(R) = \{0\}$.

Step 3: $\mathrm{ML}(B) = R$. Let $E \in \mathrm{LND}(B)$. By Proposition 3.2, $E|_R \in \mathrm{LND}(R) = \{0\}$, so $R \subseteq \ker E$. Hence $R \subseteq \mathrm{ML}(B) \subseteq R$.

Step 4: $\ker E = R$ *for every nonzero* $E \in \mathrm{LND}(B)$. By Step 3, $R \subseteq \ker E$, and by Lemma 2.4, $\mathrm{tr.deg}_k \ker E \leq n - 1 = \mathrm{tr.deg}_k R$. As in Step 1, every element of $\ker E$ is algebraic over R and hence lies in R. □

Step 4 says that B is semi-rigid: all nonzero locally nilpotent derivations have the same kernel.

Proof of Theorem C. Let $n = \mathrm{tr.deg}_k B$. Fix $D \in \mathrm{LND}^*(B)$ with slice s and $A = \ker D$. Since s is transcendental over k by Lemma 2.5, $n \geq 1$, and $\mathrm{tr.deg}_k A = n - 1 \leq 2$. As $R \subseteq A$, $\mathrm{tr.deg}_k R \leq n - 1$. If $\mathrm{tr.deg}_k R \leq 1$, Theorem A applies. The only remaining case is $n = 3$ and $\mathrm{tr.deg}_k R = 2 = n - 1$, where Theorem B applies. □

Proof of Corollary D. Let $\mathrm{tr.deg}_k B = n$, $\mathrm{ML}(B) = k$ and $k \neq R \neq B$. Since $R \neq B$, $\mathrm{LND}^*(B) \neq \emptyset$ and $\mathrm{tr.deg}_k R \leq n - 1$ by Lemma 2.5. If $\mathrm{tr.deg}_k R \leq 1$ or $\mathrm{tr.deg}_k R = n - 1$, Theorems A and B give $R = \mathrm{ML}(B) = k$, which is excluded. Hence $2 \leq \mathrm{tr.deg}_k R \leq n - 2$. For $n = 3$ this range is empty, and since $\mathrm{tr.deg}_k R = 1$ forces $R \neq k$ and $R \neq B$, no three-dimensional B has $\mathrm{ML}(B) = k$ and $\mathrm{tr.deg}_k R = 1$.

For the equivalence, let $\mathrm{tr.deg}_k B = 3$. If (III) holds, that is $\mathrm{ML}(B) = k$ and $R \neq B$, the case $R \neq k$ has just been excluded, so $R = k$, which is (II). If (II) holds, then $R = k \neq B$ because $\mathrm{tr.deg}_k B = 3$, and $k \subseteq \mathrm{ML}(B) \subseteq R = k$, which is (III). □

Remark 4.1. The implication (III)⇒(I) of [6] is false without its hypotheses: for $k = \mathbb{R}$ the ring $B = \mathbb{R}[X, Y, Z]/(XY - Z^2 - 1)\,[T]$ satisfies $\mathrm{ML}^*(B) = \mathbb{R}$ but is not $\mathbb{R}^{[3]}$ [6]. Corollary D shows that the failure happens only between (I) and the other two conditions.

5. Examples over the real numbers

The examples in this section show that ML and ML* do not commute with extension of the base field, so that results over $\mathbb{C}$ cannot simply be descended to $\mathbb{R}$. They also show that the hypotheses of Theorems A and B occur for rings that are not polynomial rings over k.

Let $S = \mathbb{R}[x, y, z] = \mathbb{R}[X, Y, Z]/(X^2 + Y^2 + Z^2 - 1)$ be the coordinate ring of the real two-sphere, and $S_{\mathbb{C}} = S \otimes_{\mathbb{R}} \mathbb{C}$. The polynomial $X^2 + Y^2 + Z^2 - 1$ is irreducible over $\mathbb{C}$, so S and $S_{\mathbb{C}}$ are domains of transcendence degree two.[4]

Proposition 5.1. (a) *S is rigid over $\mathbb{R}$, so* $\mathrm{ML}(S) = \mathrm{ML}^*(S) = S$.
(b) $\mathrm{ML}(S[T]) = \mathrm{ML}^*(S[T]) = S$, *which has transcendence degree two over* $\mathbb{R}$.
(c) $\mathrm{ML}(S_{\mathbb{C}}) = \mathbb{C}$, *and* $\mathrm{ML}(S_{\mathbb{C}}[T]) = \mathrm{ML}^*(S_{\mathbb{C}}[T]) = \mathbb{C}$.

[4]If it factored into two linear polynomials, its homogenisation $X^2 + Y^2 + Z^2 - W^2$ would be a product of two linear forms; but a product of two linear forms is a quadratic form of rank at most two, and this one has rank four. The transcendence degree is two because x, y are algebraically independent (no nonzero polynomial in X, Y alone lies in the ideal) and z is algebraic over them.

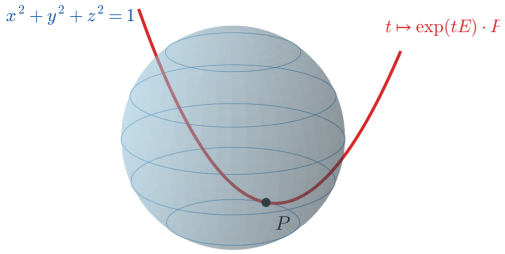


FIGURE 3. The proof of Proposition 5.1(a). For a real point P the orbit $t \mapsto \exp(tE)\cdot P$ is a polynomial curve which must stay on the sphere; a nonconstant real polynomial curve is unbounded, as drawn, so the orbit is a single point.

Proof. (a) A point $P = (p_1, p_2, p_3) \in \mathbb{R}^3$ with $p_1^2 + p_2^2 + p_3^2 = 1$ defines an $\mathbb{R}$-algebra homomorphism $\mathrm{ev}_P \colon S \to \mathbb{R}$. Let $E \in \mathrm{LND}(S)$. For $f \in S$ the function $t \mapsto \mathrm{ev}_P(\exp(tE)f) = \sum_i t^i \,\mathrm{ev}_P(E^i f)/i!$ is a real polynomial in t. Let $X(t), Y(t), Z(t)$ be these polynomials for $f = x, y, z$. Since $\exp(tE)$ is an algebra homomorphism (Lemma 2.2),

$$X(t)^2 + Y(t)^2 + Z(t)^2 = \mathrm{ev}_P\big(\exp(tE)(x^2+y^2+z^2)\big) = \mathrm{ev}_P(1) = 1$$

for all $t \in \mathbb{R}$. If the largest of the degrees of X, Y, Z were $d \geq 1$, the coefficient of t^{2d} on the left would be the sum of the squares of the leading coefficients of those polynomials of degree d, which is positive. So X, Y, Z are constant, and comparing the coefficients of t gives $\mathrm{ev}_P(Ex) = \mathrm{ev}_P(Ey) = \mathrm{ev}_P(Ez) = 0$.

Thus Ex, Ey, Ez vanish at every real point of the sphere. We show that such an element $f \in S$ is zero. Dividing by the monic polynomial $Z^2 - (1 - X^2 - Y^2)$ in Z, write $f = a(x,y) + z\,b(x,y)$ with $a, b \in \mathbb{R}[X, Y]$. For (p_1, p_2) in the open unit disc, both $(p_1, p_2, \pm\sqrt{1 - p_1^2 - p_2^2})$ are real points, so $a(p_1, p_2) \pm r\, b(p_1, p_2) = 0$ with $r = \sqrt{1 - p_1^2 - p_2^2} > 0$. Hence a and b vanish on the open disc, so $a = b = 0$ as polynomials, and $f = 0$. Therefore E kills the generators x, y, z, so $E = 0$. Since $\mathrm{LND}(S) = \{0\}$, also $\mathrm{LND}^*(S) = \emptyset$ and $\mathrm{ML}^*(S) = S$.

(b) The derivative ∂_T has the slice T and kernel S, so $\mathrm{ML}(S[T]) \subseteq \mathrm{ML}^*(S[T]) \subseteq S$. Conversely, let $E \in \mathrm{LND}(S[T])$. The argument of (a), applied at the real points (P, τ) with P on the sphere and $\tau \in \mathbb{R}$, shows that Ex, Ey, Ez vanish at all of them. An element $\sum_j f_j T^j$ of $S[T]$ ($f_j \in S$) vanishing at all these points has $f_j(P) = 0$ for all j and all P, since a real polynomial in τ with infinitely many roots is zero; so each $f_j = 0$ by (a). Hence E kills S, and $S \subseteq \mathrm{ML}(S[T])$. The same conclusion follows from Theorem B, once one knows that $\mathrm{tr.deg}_k \mathrm{ML}^*(S[T]) = 2$.

(c) Put $u = x + iy$ and $v = x - iy$ in $S_{\mathbb{C}}$. Then $uv = x^2 + y^2 = 1 - z^2$, and $S_{\mathbb{C}} = \mathbb{C}[u, v, z]$ with the single relation $uv + z^2 - 1 = 0$. Define a $\mathbb{C}$-derivation D_1 by $D_1(u) = 0$, $D_1(z) = u$, $D_1(v) = -2z$. It is well defined, since $D_1(uv + z^2 - 1) = u(-2z) + 2zu = 0$. It is locally nilpotent

by Lemma 2.1, since $D_1^2(z) = 0$ and $D_1^3(v) = -2D_1^2(z) = 0$. Symmetrically, $D_2(v) = 0$, $D_2(z) = v$, $D_2(u) = -2z$ defines $D_2 \in \mathrm{LND}(S_{\mathbb{C}})$.

The localisation $S_{\mathbb{C}}[u^{-1}]$ equals $\mathbb{C}[u^{\pm 1}, z]$, because $v = (1 - z^2)/u$. Here u and z are algebraically independent, since they generate a field of transcendence degree two. Extend D_1 to $S_{\mathbb{C}}[u^{-1}]$ by $D_1(b/u^m) = D_1(b)/u^m$, which is possible as $D_1(u) = 0$. On $\mathbb{C}[u^{\pm 1}][z]$ it acts as $u\,\partial/\partial z$, whose kernel is $\mathbb{C}[u^{\pm 1}]$. Hence $\ker D_1 \subseteq \mathbb{C}(u)$, and in the same way $\ker D_2 \subseteq \mathbb{C}(v)$. The elements u, v are algebraically independent, because z is algebraic over $\mathbb{C}(u, v)$ through $z^2 = 1 - uv$. In the field $\mathbb{C}(u, v)$ one has $\mathbb{C}(u) \cap \mathbb{C}(v) = \mathbb{C}$.[5] So $\mathrm{ML}(S_{\mathbb{C}}) \subseteq \ker D_1 \cap \ker D_2 = \mathbb{C}$.

Extend D_1 and D_2 to $S_{\mathbb{C}}[T]$ by $D_i(T) = 0$. Each commutes with ∂_T, so $D_i + \partial_T$ is locally nilpotent by Lemma 2.6, and it has the slice T. On $S_{\mathbb{C}}$ it agrees with D_i. Using also ∂_T, whose kernel is $S_{\mathbb{C}}$,

$$\mathrm{ML}^*(S_{\mathbb{C}}[T]) \subseteq S_{\mathbb{C}} \cap \ker(D_1 + \partial_T) \cap \ker(D_2 + \partial_T) = \ker D_1 \cap \ker D_2 = \mathbb{C}.$$

Since $\mathrm{ML} \subseteq \mathrm{ML}^*$ and both contain $\mathbb{C}$, both equal $\mathbb{C}$. □

So the real ring $S[T]$ has $\mathrm{ML} = \mathrm{ML}^* = S$, of transcendence degree two, while after extension to $\mathbb{C}$ both invariants become $\mathbb{C}$. The equality $\mathrm{ML} = \mathrm{ML}^*$ over $\mathbb{C}$, which is part of [9, Theorem 1], therefore carries no information about the two real invariants: both change. The hyperboloid used by Dasgupta and Gupta behaves in the opposite way, with ML^* already trivial over $\mathbb{R}$ [6].

Example 5.2. Let $C = \mathbb{R}[x, y]/(x^2 + y^2 - 1)$ be the coordinate ring of the real circle and $B = C[T_1, T_2]$. The argument of Proposition 5.1(a), with $X(t)^2 + Y(t)^2 = 1$, shows that C is rigid and that every locally nilpotent derivation of B kills C. The derivations ∂_{T_1} and ∂_{T_2} have slices and $\ker \partial_{T_1} \cap \ker \partial_{T_2} = C$. Hence $\mathrm{ML}(B) = \mathrm{ML}^*(B) = C$, of transcendence degree one, as Theorem A predicts.

6. Remarks on the open cases

Higher dimension. Let $\mathrm{tr.deg}_k B = n \geq 4$ and $2 \leq \mathrm{tr.deg}_k \mathrm{ML}^*(B) \leq n - 2$. The proof of Theorem A still gives the following: if $E \in \mathrm{LND}(B)$ does not kill $R = \mathrm{ML}^*(B)$, then $K = \ker(E|_R)$ contains elements transcendental over k, and no local slice $g \in R$ of $E|_R$ has $E(g)$ invertible in R. Lemma 3.3 gives $R \subseteq \mathrm{ML}(A)$ for every slice kernel A. For $n = 4$ and $\mathrm{tr.deg}_k R = 2$ this says that R is a two-dimensional subring of the three-dimensional ring A killed by every locally nilpotent derivation of A. Over an algebraically closed field the equality $\mathrm{ML}(B) = \mathrm{ML}^*(B)$ is known in all these cases [9]. We do not know whether it holds over $\mathbb{R}$ or $\mathbb{Q}$, and the methods above do not decide it.

Positive characteristic. In characteristic $p > 0$ locally nilpotent derivations are replaced by exponential maps, and ML and ML^* are defined in the same way. Sai Krishna asked whether $\mathrm{ML}^*(B) \neq B$ implies $\mathrm{ML}^*(B) = \mathrm{ML}(B)$ for affine domains over algebraically closed fields of arbitrary characteristic, and answered it in some cases [13]. Our proofs use characteristic zero in Lemma 2.3, where the derivative P' must be nonzero, and in the division by $i!$. They do not apply to that question.

[5] If $p(u)/q(u) = r(v)/s(v)$ with coprime p, q and coprime r, s, then $p(u)s(v) = q(u)r(v)$ in the polynomial ring $\mathbb{C}[u, v]$. Since $q(u)$ is coprime to $p(u)$, it divides $s(v)$, so q is constant; likewise s is constant, and then $p(u)$ is a constant multiple of $r(v)$, so both are constant.

Declarations

Data availability. No data were generated or analysed. The LaTeX source of this paper and the script that draws its figures are archived with this paper's release on Zenodo.

Funding. The author received no funding for this work.

Conflict of interest. The author declares no conflict of interest.

Ethics statement. This work involves no human or animal subjects.

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